DSAN 6725 · FINAL PROJECT

AN AGENTIC AI SYSTEM

Sage

A meal planning app that integrates LLMs into manual trackers.

Tyler Blue · Andrew Moy

FINAL PROJECT PRESENTATION

DSAN 6725 · SPRING 2026

APRIL 24, 2026

The current state of meal planning tools

Most digital meal-planning tool falls into one of two categories

MANUAL TRACKING

01

High cognitive overhead

Every meal and ingredient manually logged by hand. Accurate, but requires user to be disciplined.

MYFITNESSPAL | MEALIME | EATTHISMUCH

FREE-FORM LLM

02

Fun but unreliable

Fluent natural language, but hallucinates meals, invents macros, and no structured memory outside of context window.

CHATGPT DIY | GENERIC AI APPS

→ Our goal: an app you can interact with through natural language, but uses accurate nutritional info and persistent memory.

THE THESIS

An LLM based system that keeps track of your goals and profile, and automatically builds around them

01

Multi-agent pipeline

Orchestrator + Intake + Planning agents with narrow, well-defined responsibilities.

02

Persistent memory via Mem0

User profile survives across sessions. The system remembers goals, restrictions, and preferences.

03

Deterministic retrieval + audit

Meals come from a curated 751-recipe corpus. A deterministic audit layer validates every plan.

FROM RELATED WORK

~35%

Five leading LLMs undercalculated daily energy intake by ~35% (≈ 695 kcal) versus registered-dietitian reference plans.

BILEN ET AL. · 2026 · N = 5 MODELS

23 studies

A systematic review flagged safety concerns, hallucination risks, and limited real-world implementation across the ChatGPT meal planning literature.

GUO ET AL. · 2025 · SYSTEMATIC REVIEW

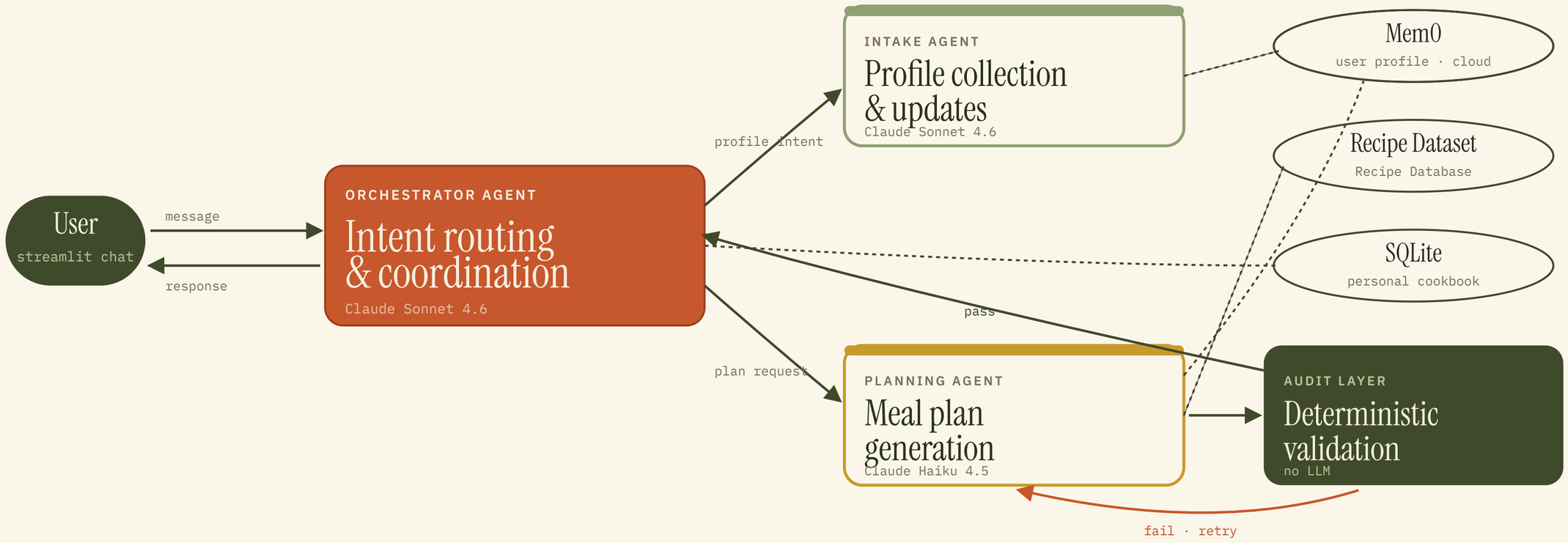
DESIGN IMPLICATION

Don't let the LLM invent meals. Instead, have it assemble plans from a recipe corpus with pre-validated nutrition facts.

THE BIG PICTURE

System Architecture

STRANDS SDK · MEM0 · SQLITE



SPECIALIZATION > GENERALIZATION

Agent Breakdown

01 · USER-FACING

Orchestrator

Claude Sonnet 4.6

Analyzes each message, routes intent to the right sub-agent, handles cookbook CRUD directly.

TOOLS

- update_user_profile
- build_meal_plan
- audit_meal_plan
- generate_shopping_list
- add_recipe_to_cookbook
- add_recipe_from_url
- edit_cookbook_recipe

02 · PROFILE

Intake Agent

Claude Sonnet 4.6

Builds and updates the user's profile through natural conversation.

TOOLS

- get_user_profile
- update_goal
- update_body_stats
- update_activity_level
- update_dietary_preferences
- update_meal_constraints
- recall_user_info

03 · PLAN GENERATION

Planning Agent

Claude Haiku 4.5

Computes targets deterministically (Mifflin-St Jeor), queries the corpus, assembles scored day or week plans.

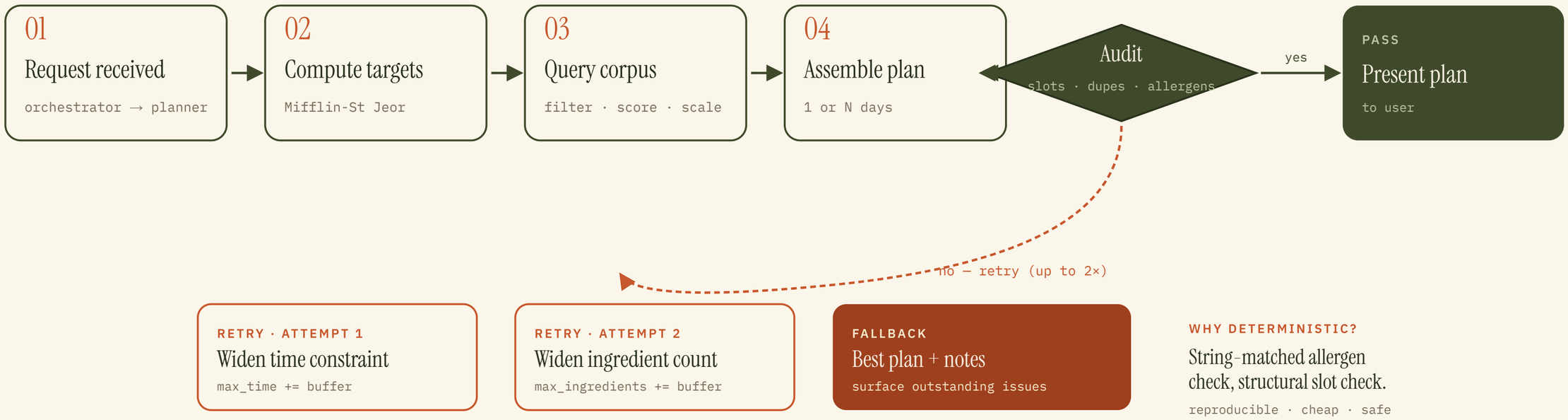
TOOLS

- get_user_profile
- estimate_calorie_target
- estimate_protein_target
- build_day_plan_tool
- build_week_plan_tool
- search_meals_tool

OUR SAFETY MECHANISM

Plan-Audit Loop

No LLM in the validation path. Every plan is checked for required slots, duplicates, and allergen violations before it reaches the user.



WHY DETERMINISTIC?

String-matched allergen check, structural slot check.

reproducible · cheap · safe

EVERY CHOICE MINIMIZES ACCIDENTAL COMPLEXITY

Architecture Decisions

Strands Agent SDK

FRAMEWORK

Simplistic tool calling and agent definitions. LangChain uses chains, runnables, and callbacks that can be difficult to interpret.

chosen over · LangChain

Claude Sonnet 4.6

REASONING AGENTS

Open-ended reasoning for intent routing and ambiguous intake conversation.

used by · orchestrator, intake

Claude Haiku 4.5

STRUCTURED AGENT

Sequential tool-calls on a fixed workflow — faster and cheaper than Sonnet.

used by · planner

Mem0

MEMORY · PROFILES

Cloud vector memory keyed by user ID. Handles embeddings internally — no need to manage infrastructure.

chosen over · local vector DB

SQLite

COOKBOOK

SQL database to store recipes by user. Works for a local or production app.

not a vector store job

Streamlit

FRONTEND

Chat-style UI with minimal frontend overhead, perfect for a demo — forwards messages, renders responses.

chosen over · custom React

BRAINTRUST HARNESS · ALL DETERMINISTIC SCORERS

29 cases, 8 scorers

TEST CASES BY CATEGORY

Meal Planning	12
Dietary Restriction	6
Profile Update	4
Out of Scope	4
Shopping List	3
Total	**29**

SCORERS · ALL DETERMINISTIC

`ToolSelection`
Recall of expected tools

`Latency`
Banded response time

`NoError`
No exceptions thrown

`ScopeAwareness`
In-scope / refusal

`PlanCompleteness`
Required plan fields

`CalorieTarget`
Within ±15%

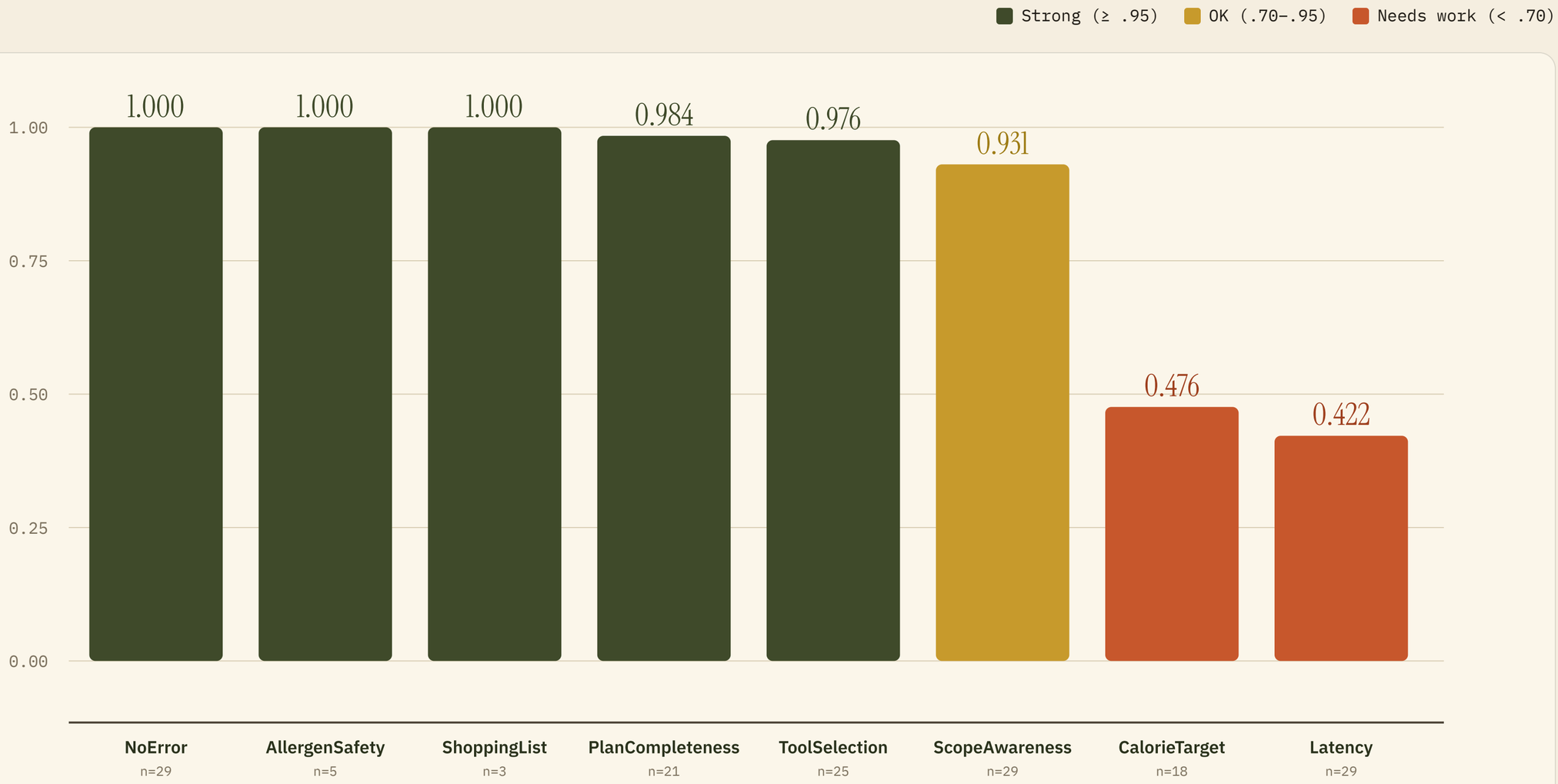
`AllergenSafety`
No restricted ingredients

`ShoppingList`
Quantities + line items

Stayed away from LLM-as-judge to keep it cheap and simple.

AVERAGE SCORE ACROSS APPLICABLE CASES

All Evaluations



THREE LOWER SCORES — AND WHAT WE'D FIX

Where the system struggles

.476

CALORIETARGET

WHY

Budget Bytes' recipes are small, 213 calories per serving on average. Muscle-gain profiles can't reach calorie targets.

FIX

Replace default cookbook with a different recipe dataset (or multiple).

.422

LATENCY

WHY

2–3 sequential Claude API calls per turn, plus Mem0 reads take over 15 seconds.

FIX

Parallel tool calls, cache Mem0 reads per session, stream output to the user.

.931

SCOPEAWARENESS

WHY

Two out-of-scope cases scored 0. The system responded correctly on manual review, it just didn't use the exact phrase the scorer expects.

FIX

Need to calibrate the scorer rather than the agents. We'd either tune the scorer's regex or add a lightweight refusal classifier.

SWITCHING TO STREAMLIT

Demo

Profile intake · Multi-day plan · Allergen case · Shopping list

01 · ONBOARD A NEW USER
02 · BUILD A 3-DAY PLAN
03 · TRIGGER AN ALLERGEN RETRY
04 · GENERATE SHOPPING LIST

WHAT WE SHOWED

Conversational Multi-Agent Meal Planning

An agentic architecture can deliver conversational UX *and* general nutritional reliability.

DSAN 6725 · APRIL 2026

Slides crafted with Claude Cowork

FUTURE WORK

Lab-validated corpus

USDA FoodData Central — fixes CalorieTarget

Grocery MCP integration

Plan → cart, end-to-end

Temporal profile

Weight & intake tracking; adaptive targets

Production deploy

Auth, hosting, polished UI

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THANK YOU · QUESTIONS?